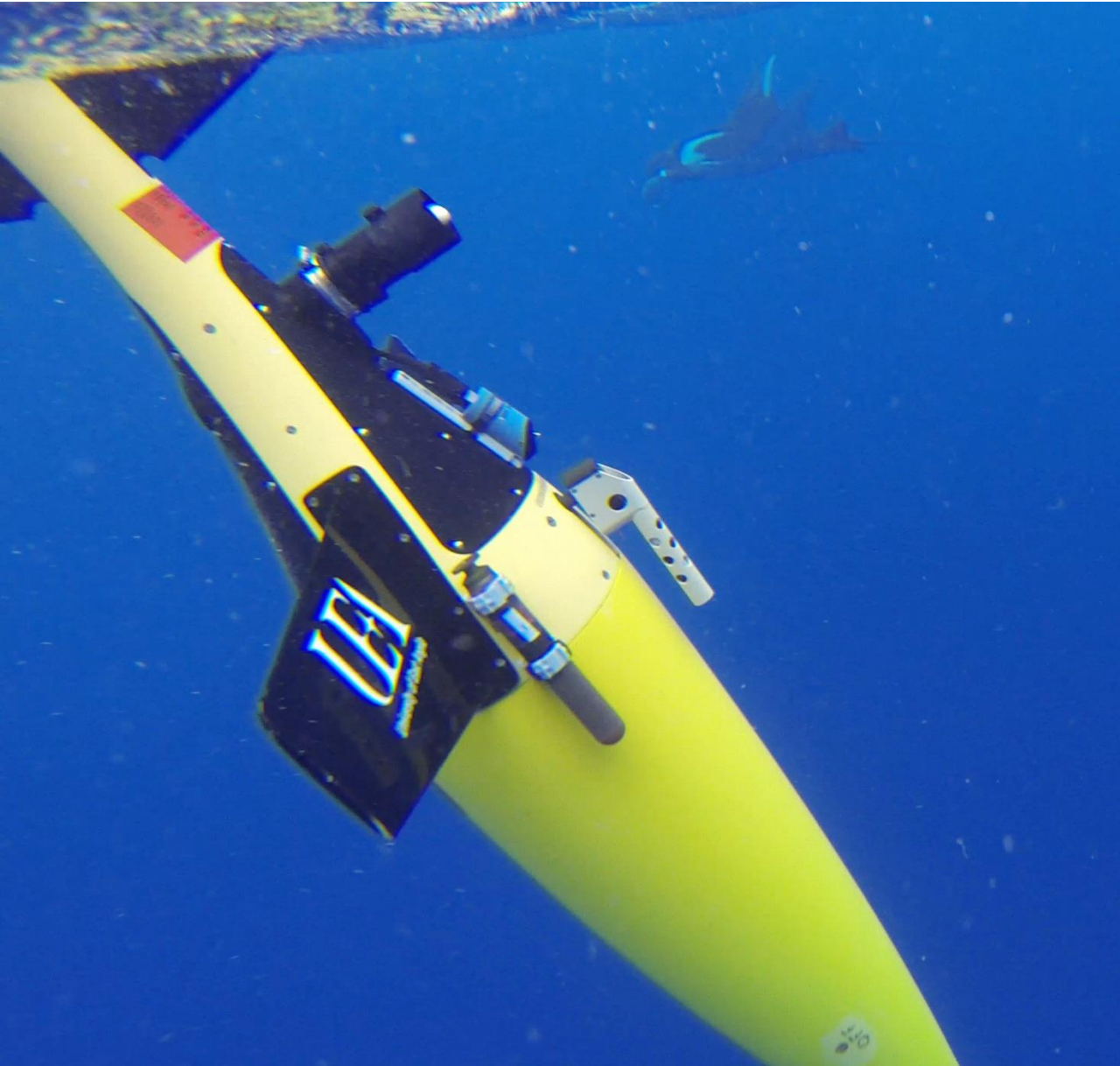


Noise in glider recordings

Pierre Cauchy



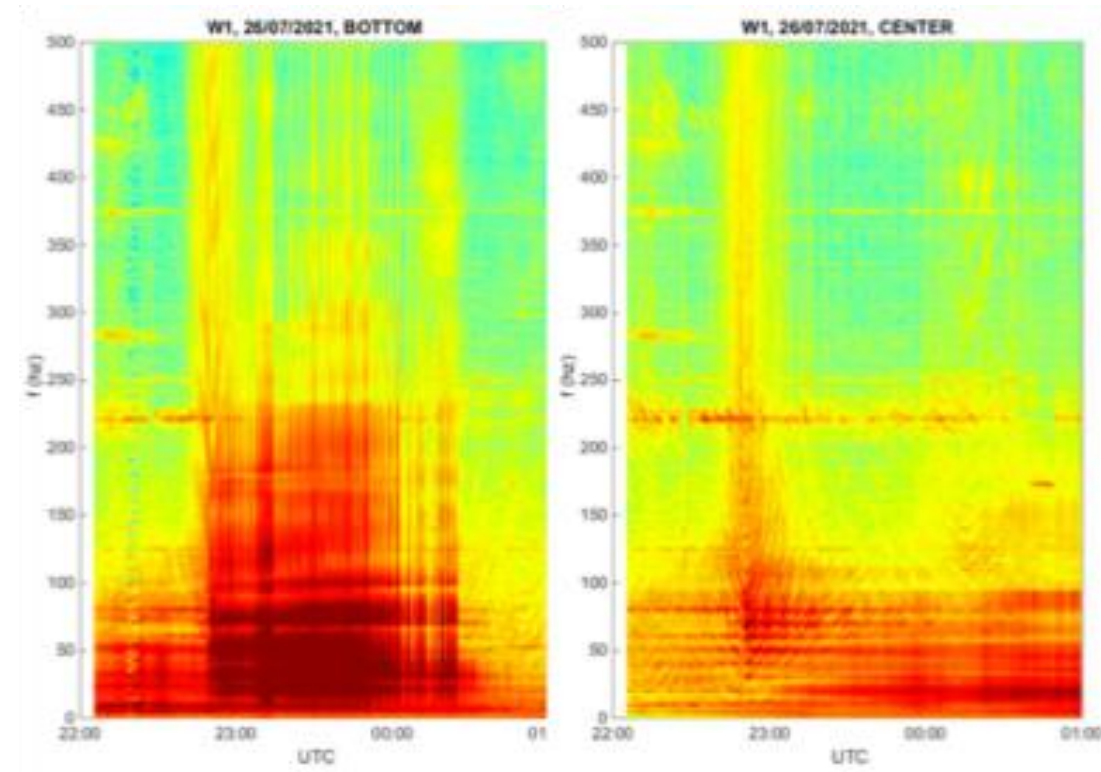
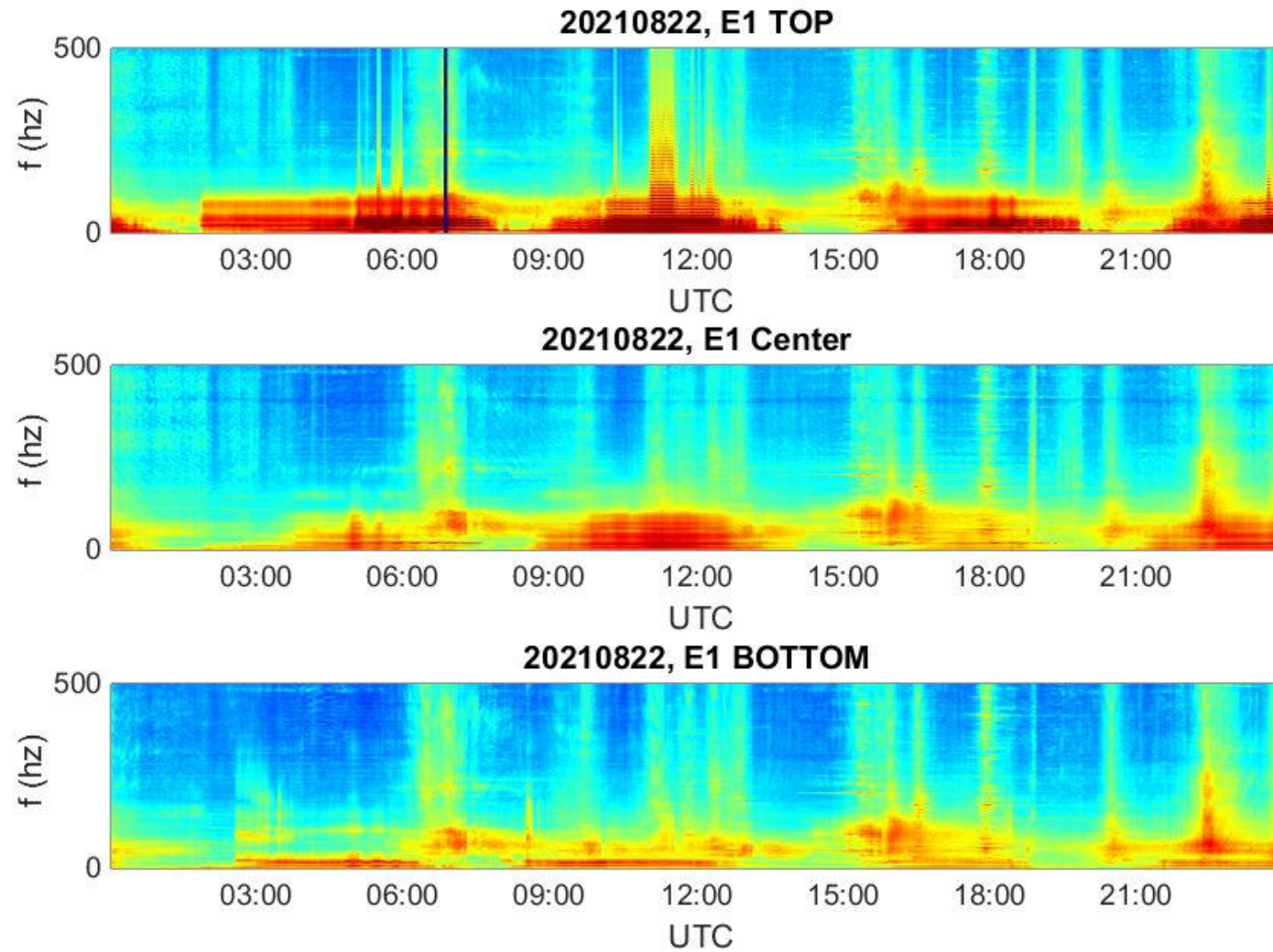
Noise sources

Workaround

Impact on observation



Noise on moored hydrophones



Strumming

Mobile parts

Current flow

Gliders are quiet

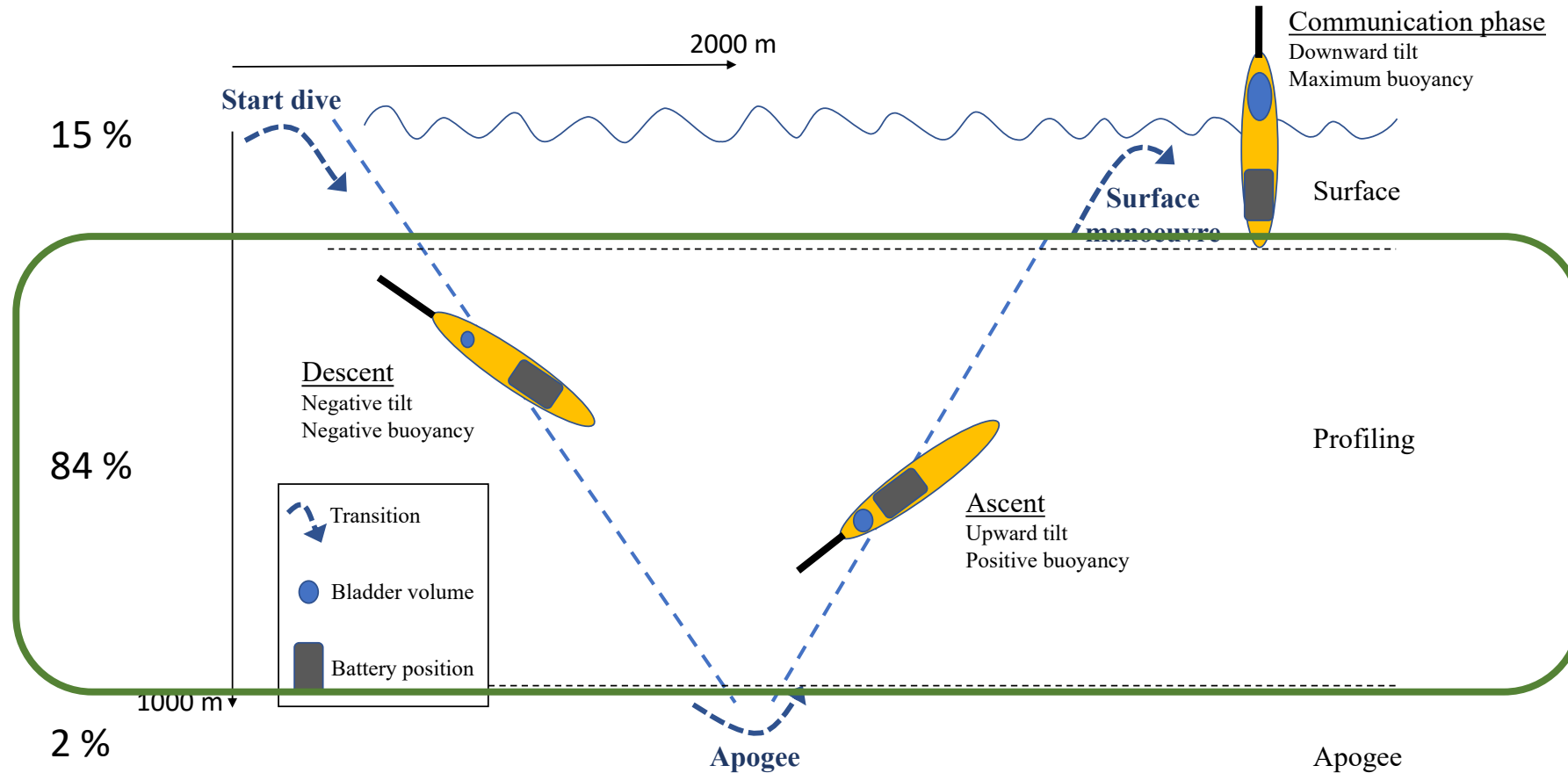
No propulsion

Rare / Intermittent operation noise

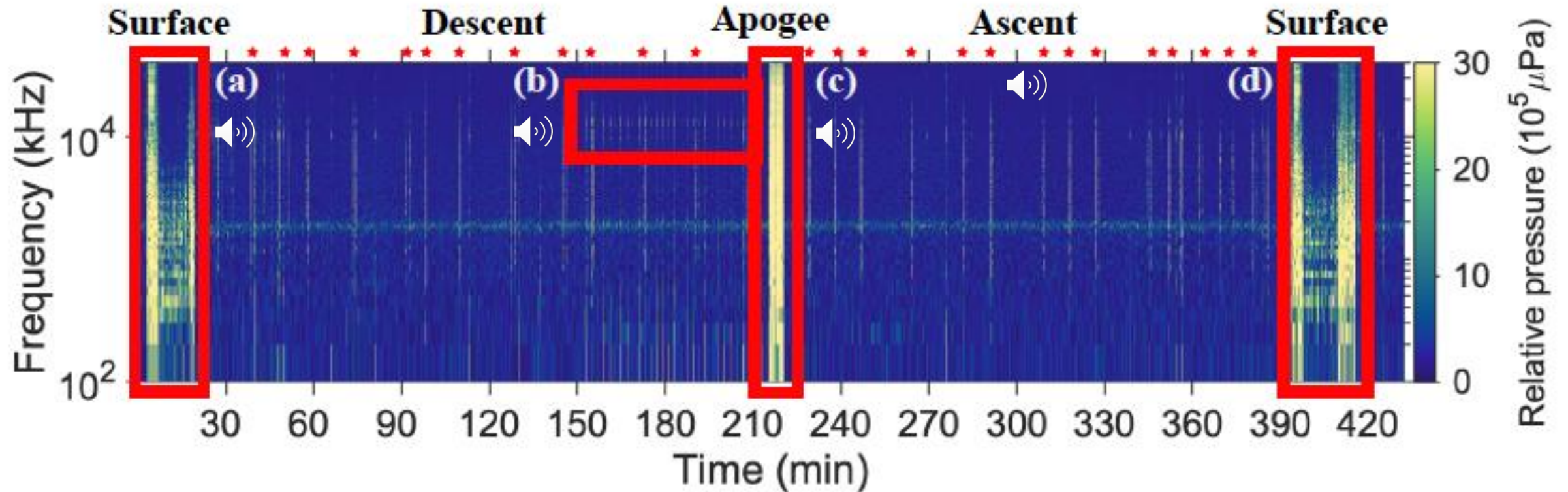
Limited flow speed

No mobile parts

Buoyancy propelled – Quiet platform

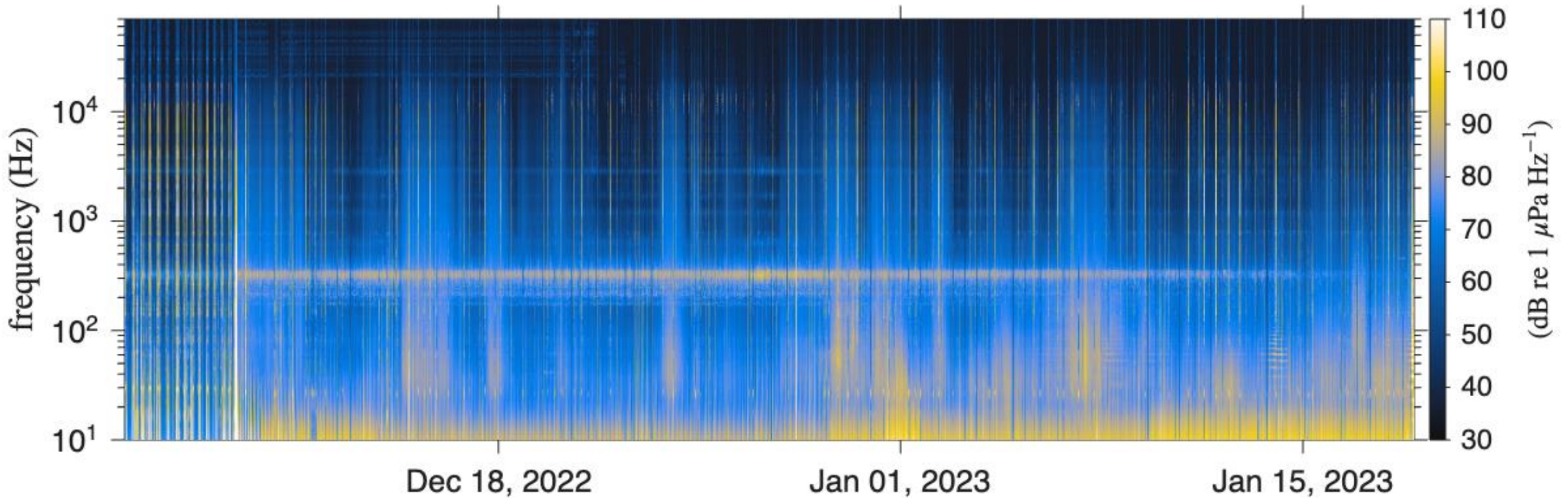


Glider navigation noise – Rare and predictable



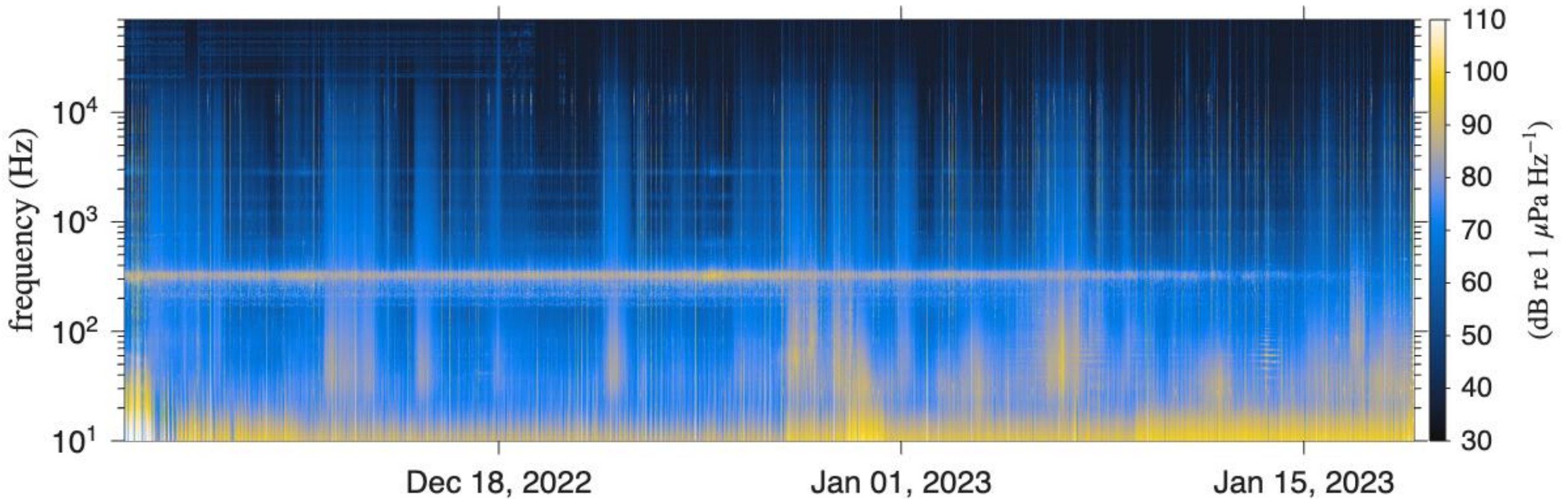
Removing glider navigation noise

Raw data
10 780 samples



Removing glider navigation noise

Buoyancy pump removed
9 021 / 10 780

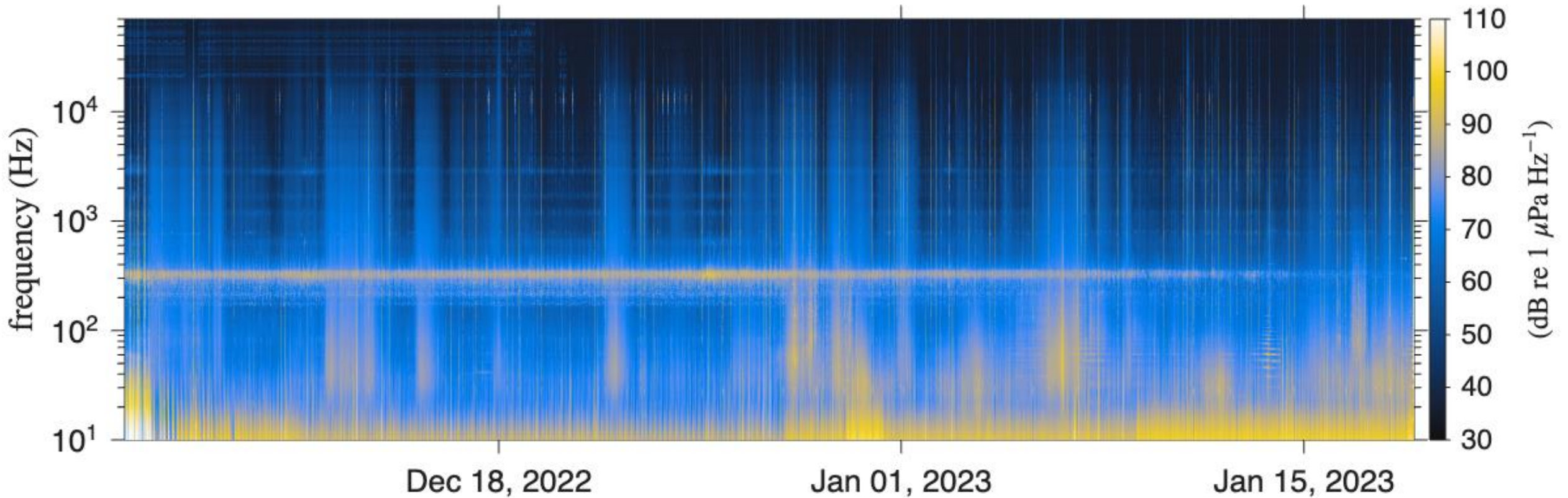


Removing glider navigation noise

Surface noise removed

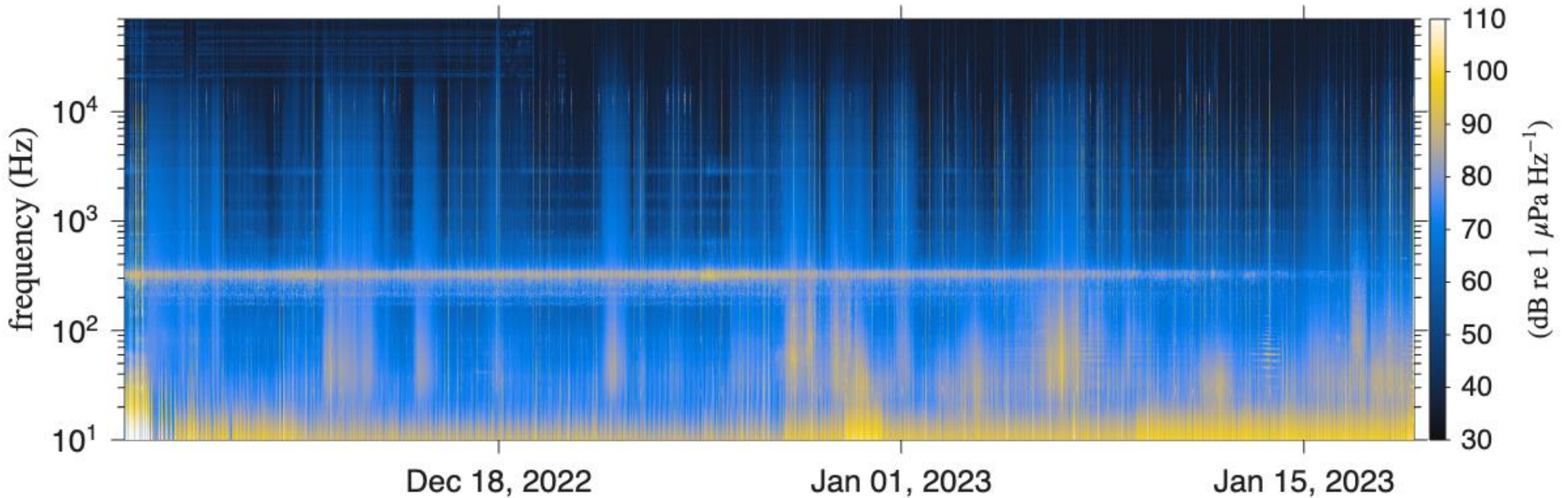
Depth > 20 m

8 834 / 10 780

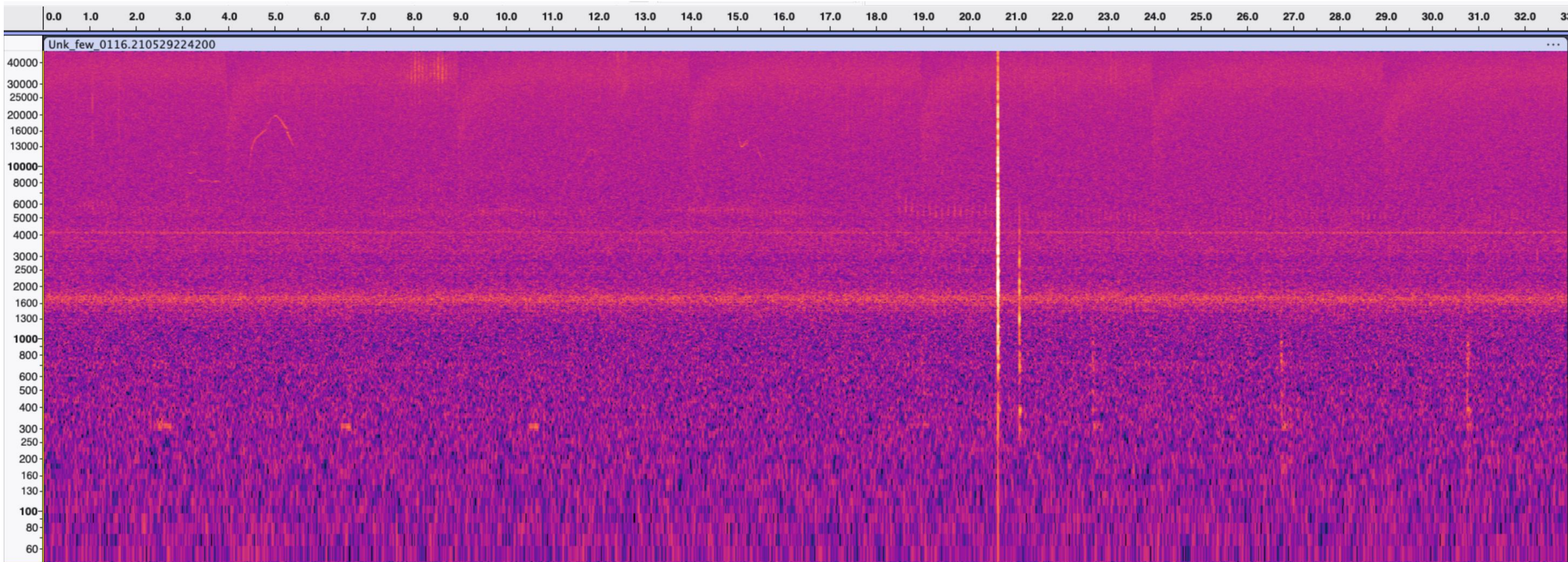


Removing glider navigation noise

Under-ice stuck removed
Vertical speed > 0
8 817 / 10 780 (82%)

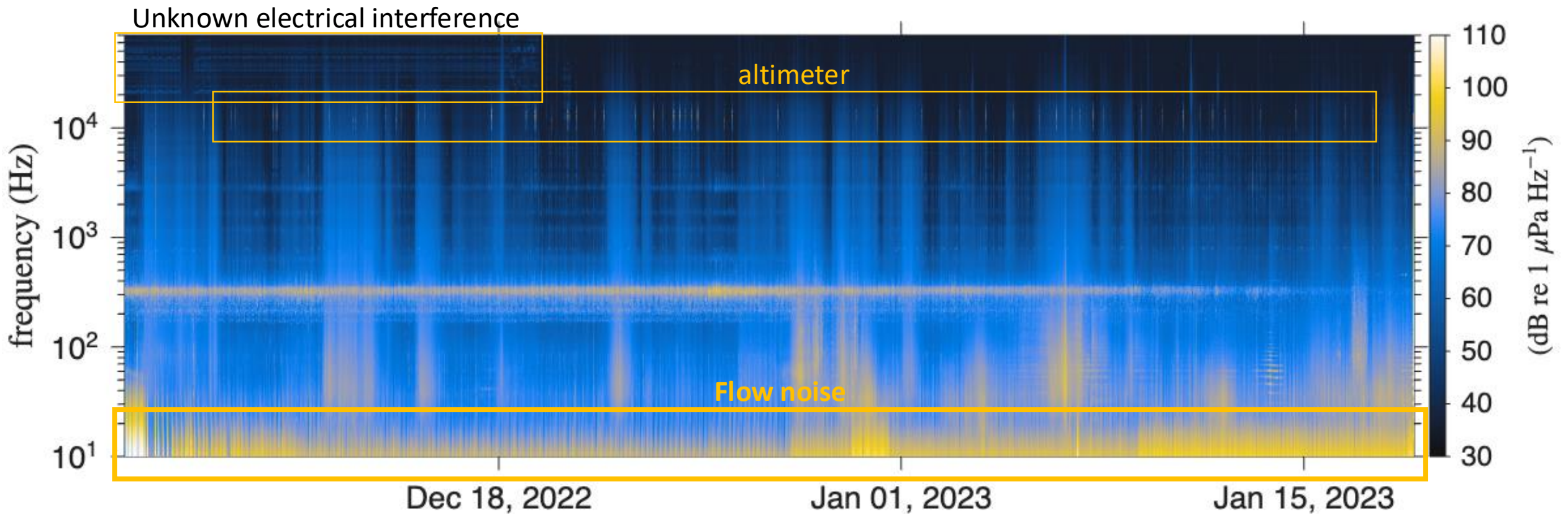


Unknown – Glider related noise

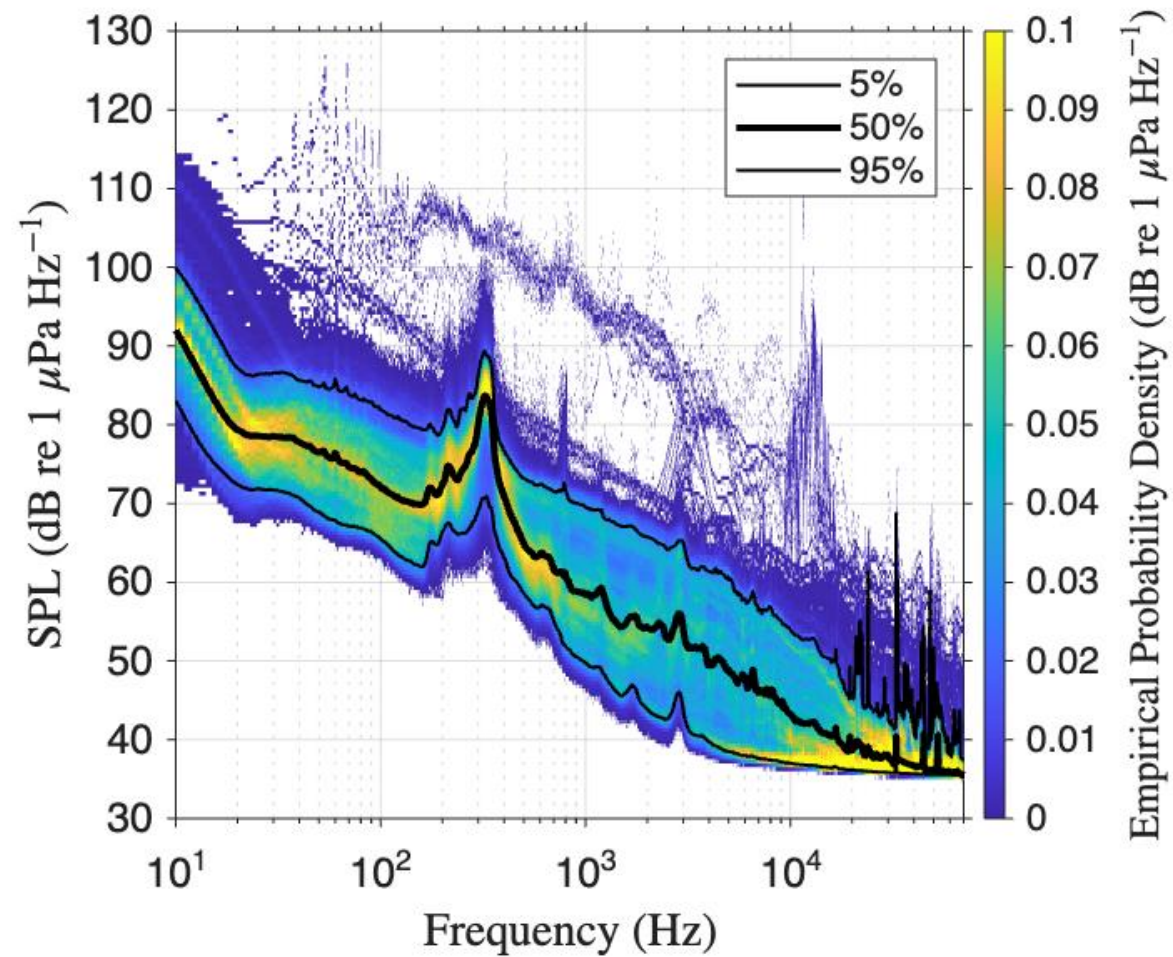
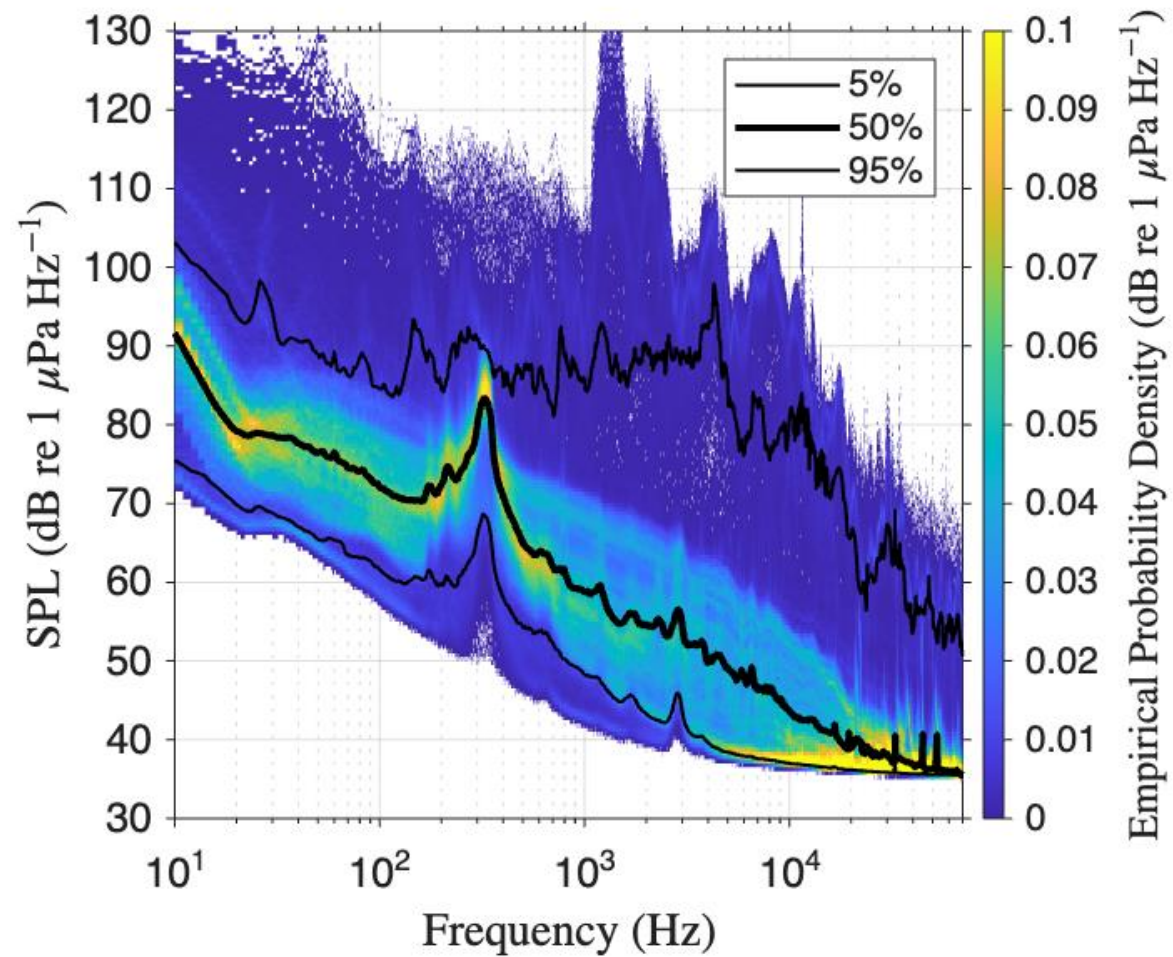


Removing glider navigation noise

Additional manual detection
8 426 / 10 780 (78%)



Before / After



Flow noise

Pseudo noise

Does not exist but is recorded
Induced by motion through water
Turbulence / pressure variations
Low frequency

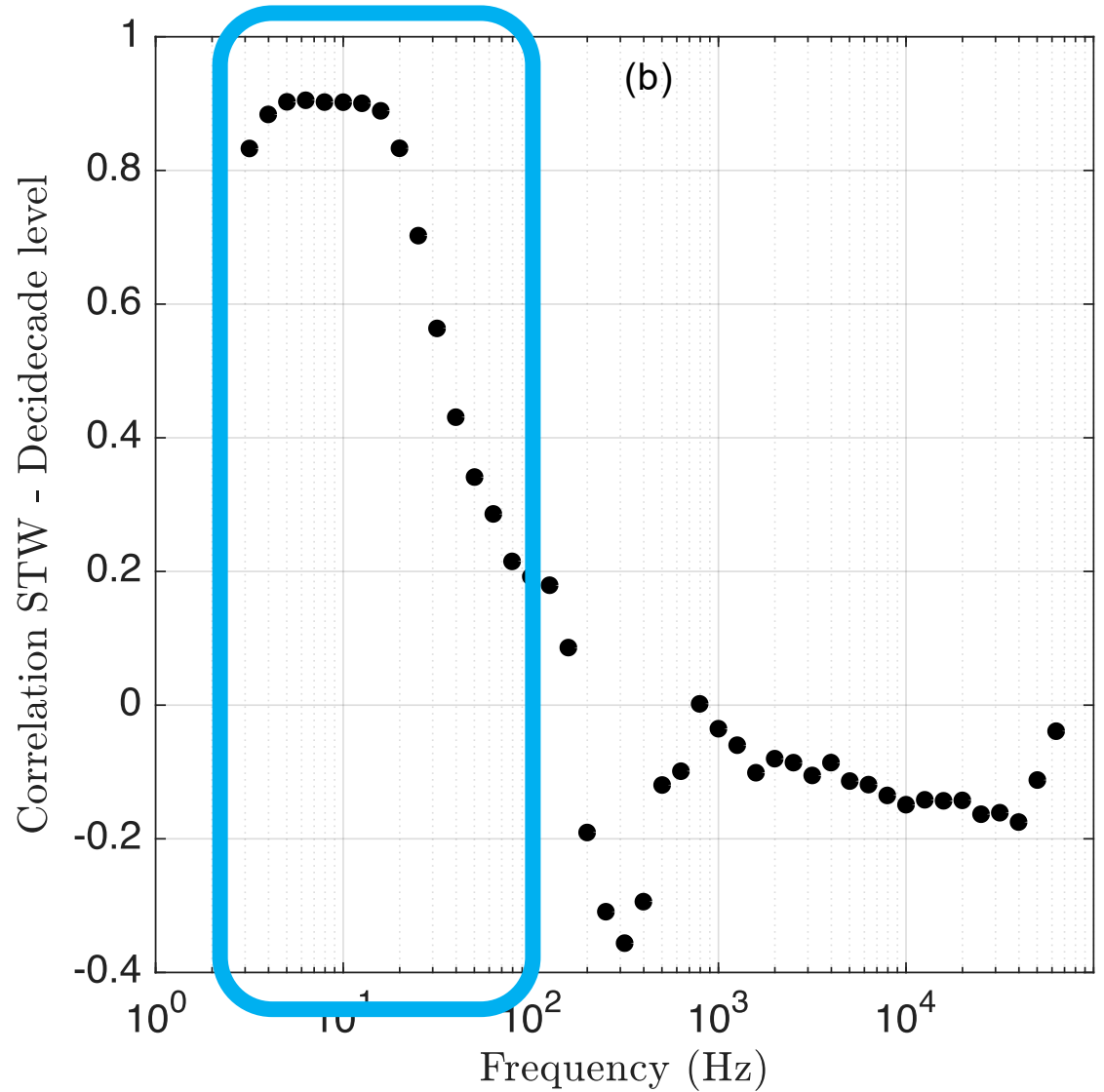
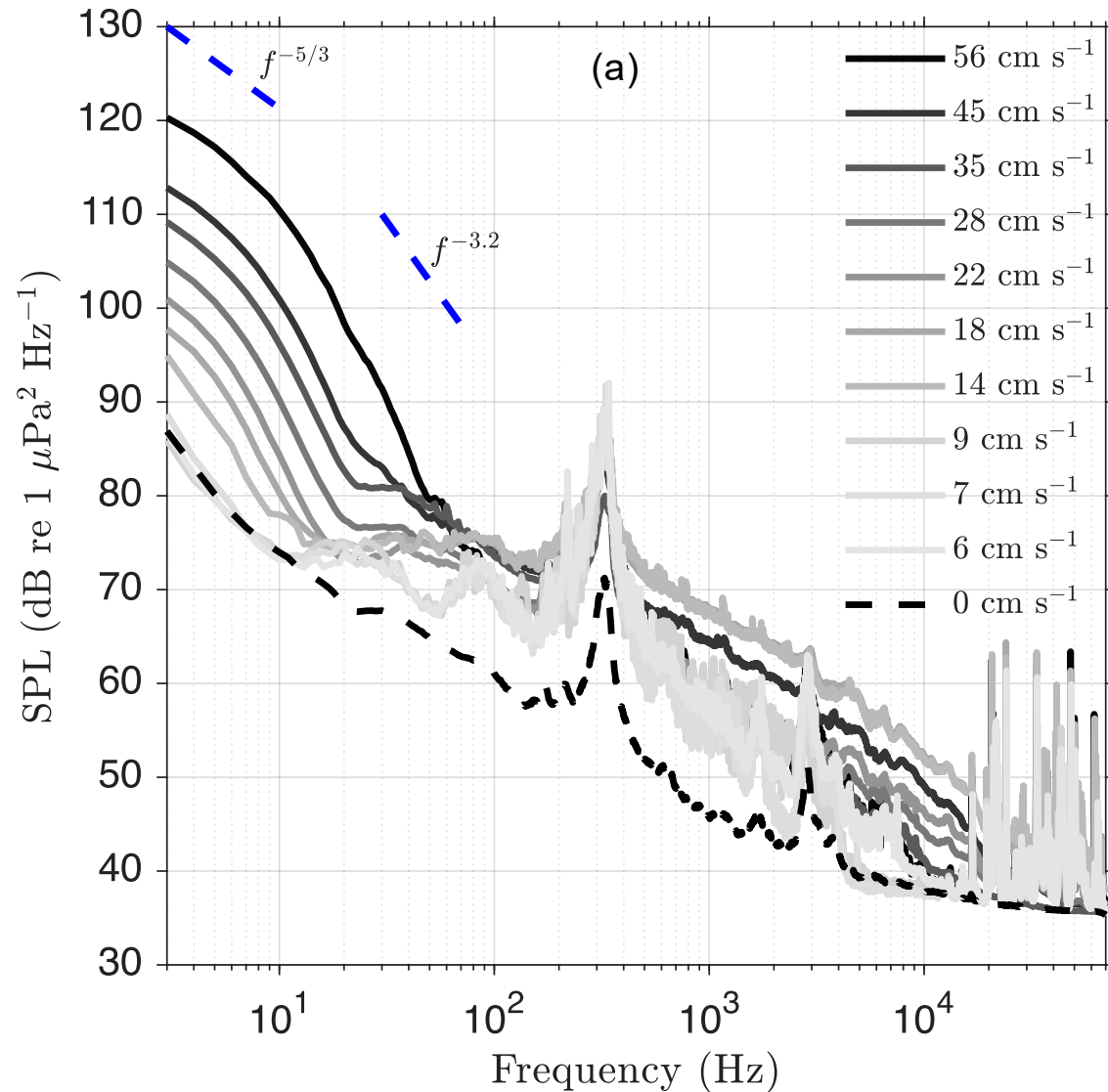
Aggravating factors

Shape	Streamlined
Speed Through Water	Slow

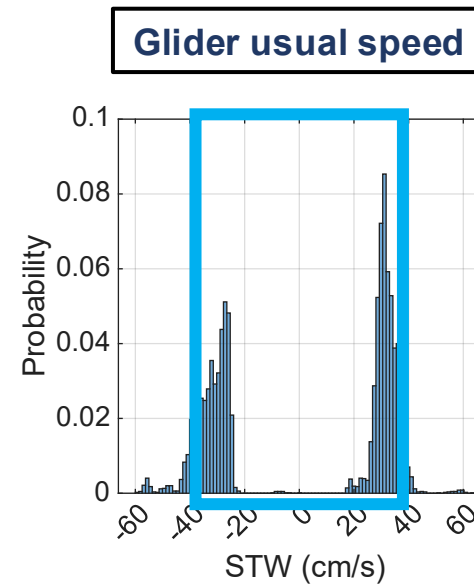
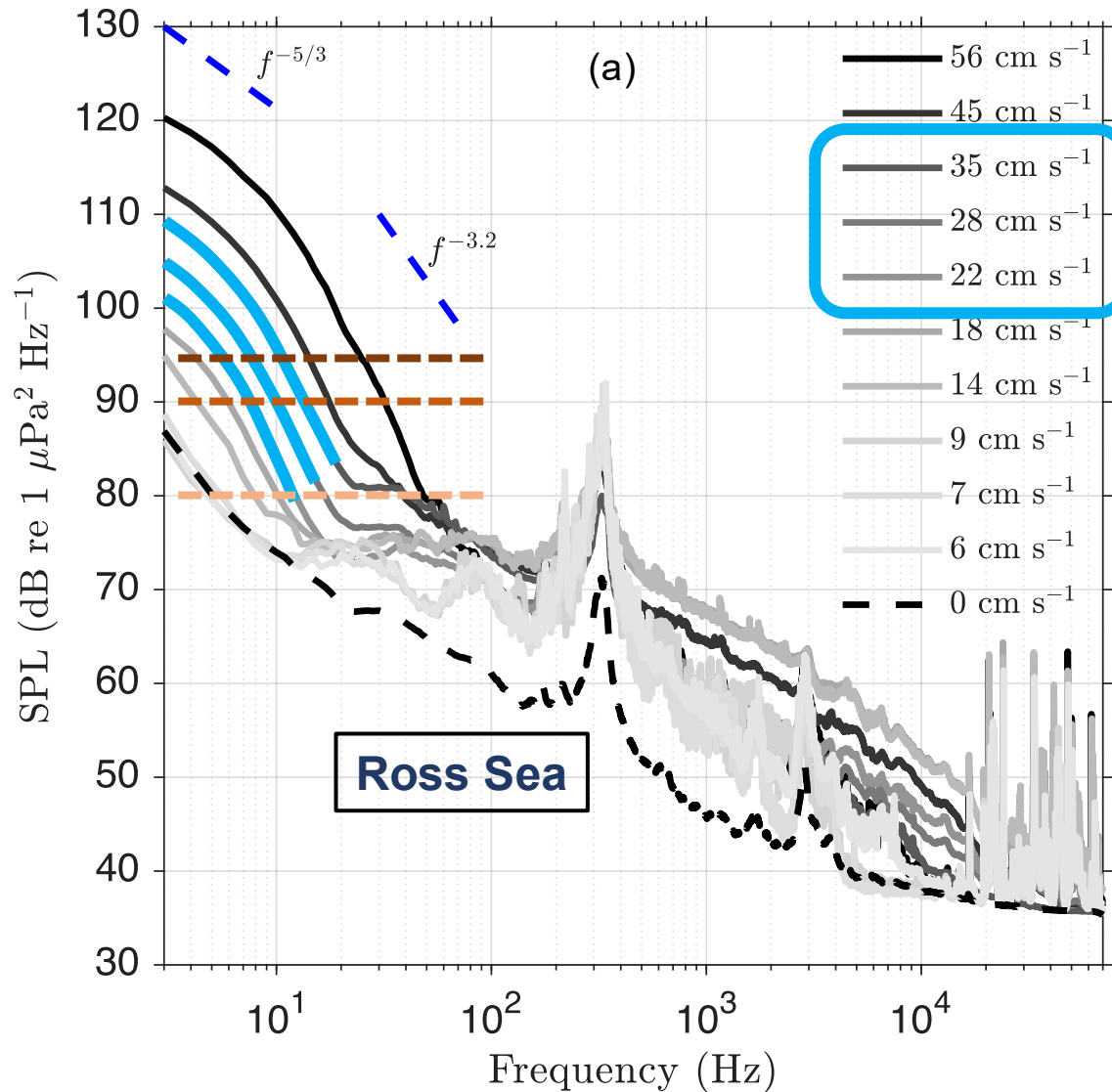
Usual flow speed

Tidal currents – up to 4 m/s
Oceanic currents – up to 1 m/s
Glider – up to 0.6 m/s

Glider-generated flow noise



Glider-generated flow noise



20 Hz usual level

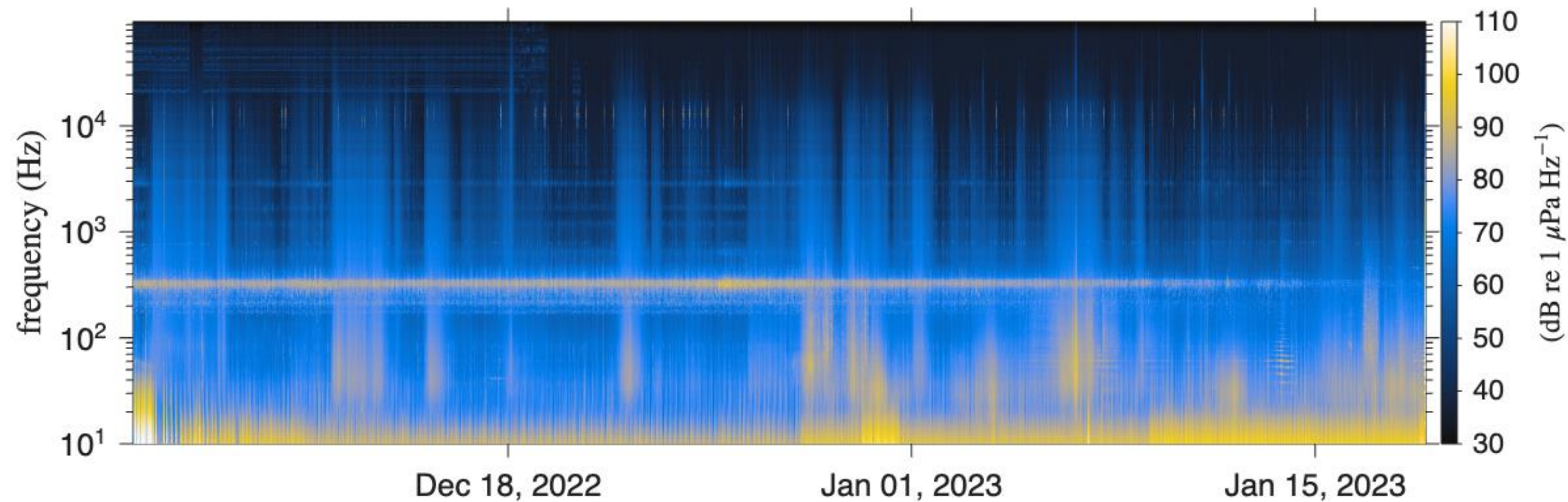
Wake Island & Perth Canyon – 80 dB

90s traffic off Oregon coast – 90 dB

2020s traffic St. Lawrence – 95 dB

Cauchy, P., (2026) Marginal effect of flow noise on Ocean Sound measurement from underwater gliders, *J. Acoust. Soc. Am.*

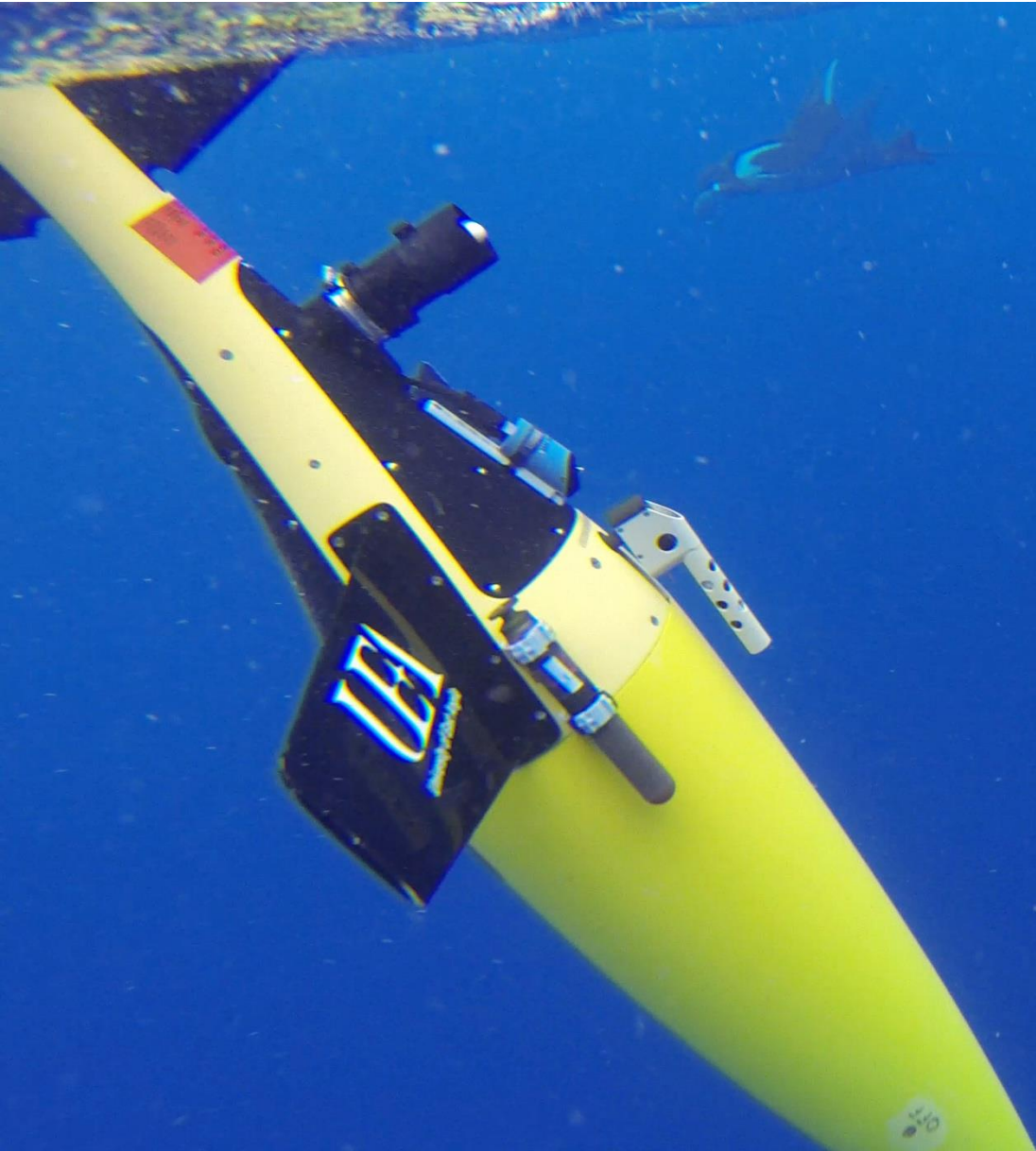
Effects of flow noise on my measurements?



	< 12 Hz	12 – 40 Hz	> 40Hz
Detectable flow noise?	Always	STW > 25 m s ⁻¹	STW > 60 cm s ⁻¹
Speed adjustment	As slow as possible	Adapt speed to ambient noise level	Don't worry

Noise in glider recordings

Pierre Cauchy



Technology

Piloting / Setup

Postprocessing

Effects on measurements

